

# Weekly Progress Report

## Amity School of Engineering & Technology

**Project Group ID - 265**

**Project Area – Data Science**  
(Machine Learning)

**B. Tech (8CSE-DS-1Y) VIII Semester**

**WPR No. 9**

**Duration: 02/02/2026 to 08/02/2026**

Students Weekly Progress Report (WPR) for Even Semester of session 2025-2026

|                                                                |                                                                                                         |
|----------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>To be filled by Students</b>                                |                                                                                                         |
| Students Name                                                  | 1. Sagar Singh Bisht<br>2. Aakarsh Pab                                                                  |
| Roll no.                                                       | 1. 056<br>2. 050                                                                                        |
| Enrollment no.                                                 | 1. A023167022056<br>2. A023167022050                                                                    |
| Project Title finalized, if Yes, give name, if NO, give reason | Machine Learning based Predictive Model for Livability using Environmental and Socioeconomic indicators |
| Synopsis submitted                                             | YES                                                                                                     |
| Literature review                                              | YES                                                                                                     |
| Technical & Economical Feasibility                             | YES                                                                                                     |
| Bill of Material                                               | NA                                                                                                      |
| Project Progress Schedule (PERT Chart)                         | NA                                                                                                      |
| Design of critical components                                  | -                                                                                                       |
| Fabrication work (give %)                                      | -                                                                                                       |
| Experimental work (give %)                                     | -                                                                                                       |
| Result and Analysis                                            | -                                                                                                       |
| Report writing                                                 | -                                                                                                       |
| Signature of students                                          |                                                                                                         |

## Work done in this week:

Worked on the web page design of the project.

Integrated tomtom API for the traffic data to enhance the project and accessing the real time data of traffic

## Plan for Next Week

- Framework Initialization: Setup the Streamlit environment and define the application architecture to ensure the backend script can be imported as a module.
- UI/UX Design: \* Construct a Sidebar Control Panel featuring st.slider widgets for each priority (Rent, Air Quality, Traffic, Amenities).
  - Implement Constraint Inputs (e.g., a numeric input for "Maximum Budget") to act as a hard filter before the similarity ranking begins.
- Dynamic Result Display: Create a "Results Gallery" using st.columns and st.metric to showcase the top 5 recommended neighborhoods with their respective similarity scores.

### 2. Integration & Real-Time Logic

- Backend-to-Frontend Bridge: Connect the Cosine Similarity function to the Streamlit session state so that the recommendations update dynamically as the user moves the sliders.
- Data Visualization: \* Integrate st.map to provide a spatial view of the recommended areas.
  - Develop a Radar Chart (using Plotly) for each recommendation to show how well the neighborhood matches the user's specific priorities (e.g., "High match on Rent, low match on Metro")
- .
- Doing the data set master and making the master data set to get the prediction more powerful and efficient

|                                                                    |                    |
|--------------------------------------------------------------------|--------------------|
| To be filled with a Guide (strike off whichever is not applicable) |                    |
| The performance of students is satisfactory                        |                    |
| The performance of students is unsatisfactory                      |                    |
| A warning to be issued to student(s) (Name)                        |                    |
| Student was not well (Name)                                        |                    |
| Date    08/02/2026                                                 | Signature of Guide |